

Section 2: Selected exercise solutions

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Problem 1. Let $c = c(t)$ be a Frenet curve in $\mathbb{R}^3$, written in an arbitrary regular parameter. Prove that its curvature and torsion are given by

$$\kappa(t) = \frac{\|\dot{c}(t) \times \ddot{c}(t)\|}{\|\dot{c}(t)\|^3}, \quad \tau(t) = \frac{\det(\dot{c}(t), \ddot{c}(t), \ddot{\ddot{c}}(t))}{\|\dot{c}(t) \times \ddot{c}(t)\|^2}.$$

For an oriented plane curve, prove that the signed curvature is

$$k(t) = \frac{\det(\dot{c}(t), \ddot{c}(t))}{\|\dot{c}(t)\|^3}.$$

Solution. Write $v = \|\dot{c}\|$. Since $\dot{c} = vT$ and $\frac{d}{dt} = v\frac{d}{ds}$, the Frenet equations give

$$\ddot{c} = \dot{v}T + v^2\kappa N, \quad \dot{c} \times \ddot{c} = v^3\kappa B.$$

Taking norms gives

$$\kappa = \frac{\|\dot{c} \times \ddot{c}\|}{\|\dot{c}\|^3}.$$

The B -component of $\ddot{c}$ is $v^3\kappa\tau B$. Consequently,

$$\det(\dot{c}, \ddot{c}, \ddot{\ddot{c}}) = v^6\kappa^2\tau, \quad \|\dot{c} \times \ddot{c}\|^2 = v^6\kappa^2.$$

Thus

$$\tau = \frac{\det(\dot{c}, \ddot{c}, \ddot{\ddot{c}})}{\|\dot{c} \times \ddot{c}\|^2}.$$

For an oriented plane curve, the same calculation gives $\det(\dot{c}, \ddot{c}) = v^3k$, which proves the final formula.

Problem 4. Show that a regular curve between two points $p, q \in \mathbb{R}^n$ with minimal length is necessarily the line segment from p to q . Hint: apply the Cauchy–Schwarz inequality to the tangent vector and the difference vector $p - q$.

Solution. Assume $p \neq q$, and set

$$u = \frac{q - p}{\|q - p\|}.$$

For any regular curve $c : [a, b] \rightarrow \mathbb{R}^n$ from p to q , Cauchy–Schwarz gives

$$\begin{aligned} L(c) &= \int_a^b \|\dot{c}(t)\| dt \\ &\geq \int_a^b \langle \dot{c}(t), u \rangle dt \\ &= \langle q - p, u \rangle = \|q - p\|. \end{aligned}$$

The line segment attains this bound.

Equality in Cauchy–Schwarz requires

$$\dot{c}(t) = \lambda(t)u, \quad \lambda(t) > 0.$$

Thus $c(t) = p + \mu(t)u$, where μ ranges from 0 to $\|q - p\|$. The image of c is therefore precisely the segment joining p to q .

Problem 5. Show that if all tangent vectors to the curve $c(t) = (3t, 3t^2, 2t^3)$ are drawn from the origin, then their endpoints lie on the surface of a circular cone whose axis is the line $x - z = y = 0$.

Solution. Draw the tangent vectors from the origin and denote their endpoints by

$$X(t) = \dot{c}(t) = (3, 6t, 6t^2).$$

Let

$$a = \frac{1}{\sqrt{2}}(1, 0, 1),$$

the unit vector along the line $x - z = y = 0$. Then

$$\langle X(t), a \rangle = \frac{3 + 6t^2}{\sqrt{2}},$$

and

$$\|X(t)\| = \sqrt{9 + 36t^2 + 36t^4} = 3(1 + 2t^2).$$

Hence

$$\frac{\langle X(t), a \rangle}{\|X(t)\|} = \frac{1}{\sqrt{2}}.$$

Every $X(t)$ makes the constant angle $\pi/4$ with the axis spanned by a . Thus the endpoints lie on the circular cone with that axis and half-angle $\pi/4$.

Problem 7. Calculate explicitly a parametrization of the plane curve whose signed curvature is $k(s) = s^{-1/2}$. Hint: use the reconstruction of a plane curve from its curvature function.

Solution. We first recall the reconstruction idea. If a unit-speed plane curve has unit tangent

$$T(s) = (\cos \theta(s), \sin \theta(s)),$$

then

$$T'(s) = \theta'(s)(-\sin \theta(s), \cos \theta(s)).$$

Thus its signed curvature is $k(s) = \theta'(s)$. Conversely, given a signed-curvature function k , we can first recover the turning angle and then the curve by two integrations:

$$\theta(s) = \theta_0 + \int_{s_0}^s k(\sigma) d\sigma, \quad c(s) = p_0 + \int_{s_0}^s (\cos \theta(\sigma), \sin \theta(\sigma)) d\sigma.$$

The choices of θ_0 and p_0 only rotate and translate the resulting curve. This is the reconstruction of a plane curve from its signed curvature.

We now apply it to $k(s) = s^{-1/2}$, for $s > 0$. Since this function is integrable at 0, we may choose the limiting initial angle $\theta(0) = 0$. Then

$$\theta(s) = \int_0^s \sigma^{-1/2} d\sigma = 2\sqrt{s},$$

and hence

$$T(s) = (\cos(2\sqrt{s}), \sin(2\sqrt{s})).$$

Choosing $c(0) = (0, 0)$, the second integration gives

$$c(s) = \int_0^s (\cos(2\sqrt{\sigma}), \sin(2\sqrt{\sigma})) d\sigma.$$

Set $u = \sqrt{\sigma}$, so that $d\sigma = 2u \, du$. The two coordinates are therefore

$$\begin{aligned} x(s) &= \int_0^{\sqrt{s}} 2u \cos(2u) \, du = \sqrt{s} \sin(2\sqrt{s}) + \frac{1}{2} \cos(2\sqrt{s}) - \frac{1}{2}, \\ y(s) &= \int_0^{\sqrt{s}} 2u \sin(2u) \, du = -\sqrt{s} \cos(2\sqrt{s}) + \frac{1}{2} \sin(2\sqrt{s}). \end{aligned}$$

Thus one reconstructed curve is

$$c(s) = \begin{pmatrix} \sqrt{s} \sin(2\sqrt{s}) + \frac{1}{2} \cos(2\sqrt{s}) - \frac{1}{2} \\ -\sqrt{s} \cos(2\sqrt{s}) + \frac{1}{2} \sin(2\sqrt{s}) \end{pmatrix}, \quad s > 0.$$

Finally, differentiating the integral expression shows that $c'(s) = T(s)$, so $\|c'(s)\| = 1$. Moreover,

$$\det(T(s), T'(s)) = \theta'(s) = s^{-1/2},$$

which verifies the prescribed signed curvature. Any other solution with the same orientation differs from this one by a rotation and a translation.

Problem 9. Let a plane curve be given in polar coordinates (r, φ) by $r = r(\varphi)$. Using the notation $r' = dr/d\varphi$, show that its length over $[\varphi_1, \varphi_2]$ is

$$L = \int_{\varphi_1}^{\varphi_2} \sqrt{(r')^2 + r^2} d\varphi,$$

and that its signed curvature is

$$k(\varphi) = \frac{2(r')^2 - rr'' + r^2}{((r')^2 + r^2)^{3/2}}.$$

Solution. Let

$$c(\varphi) = r(\varphi)e_r(\varphi),$$

where

$$e_r = (\cos \varphi, \sin \varphi), \quad e_\varphi = (-\sin \varphi, \cos \varphi).$$

Since $e'_r = e_\varphi$ and $e'_\varphi = -e_r$,

$$\dot{c} = r'e_r + re_\varphi, \quad \ddot{c} = (r'' - r)e_r + 2r'e_\varphi.$$

Therefore

$$\|\dot{c}\| = \sqrt{(r')^2 + r^2},$$

and

$$L = \int_{\varphi_1}^{\varphi_2} \sqrt{(r')^2 + r^2} d\varphi.$$

The frame (e_r, e_φ) is positively oriented, so

$$\det(\dot{c}, \ddot{c}) = 2(r')^2 - r(r'' - r) = 2(r')^2 - rr'' + r^2.$$

Using the plane-curvature formula from Problem 1,

$$k(\varphi) = \frac{2(r')^2 - rr'' + r^2}{((r')^2 + r^2)^{3/2}}.$$

Problem 10. Calculate the curvature of the curve given in polar coordinates by $r(\varphi) = a\varphi$, where $a \neq 0$ is constant. This curve is called the Archimedean spiral.

Solution. Here

$$r = a\varphi, \quad r' = a, \quad r'' = 0.$$

Substitution into the formula from Problem 9 gives

$$\begin{aligned} k(\varphi) &= \frac{2a^2 + a^2\varphi^2}{(a^2 + a^2\varphi^2)^{3/2}} \\ &= \boxed{\frac{\varphi^2 + 2}{|a|(1 + \varphi^2)^{3/2}}}. \end{aligned}$$

This assumes $a \neq 0$. For the usual convention $a > 0$, the absolute value may be omitted.

Problem 12. In plane polar coordinates (r, φ) , consider the curve given by $r = \cos(2\varphi)$, $0 \leq \varphi \leq 2\pi$. Determine whether this curve is regular and, if so, calculate its rotation index and total curvature.

Solution. Since

$$r' = -2 \sin(2\varphi),$$

the speed satisfies

$$\|\dot{c}(\varphi)\|^2 = (r')^2 + r^2 = 4 \sin^2(2\varphi) + \cos^2(2\varphi) > 0.$$

Thus the curve is regular, including when $r = 0$.

In complex notation,

$$\dot{c}(\varphi) = e^{i\varphi} (r'(\varphi) + ir(\varphi)).$$

The factor $e^{i\varphi}$ winds once. The ellipse

$$r' + ir = -2 \sin(2\varphi) + i \cos(2\varphi)$$

winds twice counterclockwise about the origin. Hence

$$\boxed{U_c = 1 + 2 = 3.}$$

Moreover,

$$2(r')^2 - rr'' + r^2 = 8 \sin^2(2\varphi) + 5 \cos^2(2\varphi) > 0.$$

The signed curvature never changes sign, so

$$\boxed{\int_c k \, ds = 2\pi U_c = 6\pi.}$$

The total absolute curvature is also 6π .

Problem 16. Show that a slope line with $\tau \neq 0$ lies on a sphere if and only if

$$\kappa^2(s) = (-A^2s^2 + Bs + C)^{-1}$$

for constants A, B, C , where $A = \tau/\kappa$. Hint: use the spherical-curve criterion from Theorem 2.10(ii).

Solution of the first part. For a slope line,

$$A = \frac{\tau}{\kappa}$$

is constant. Since $\tau \neq 0$, also $A \neq 0$. The spherical-curve criterion from Theorem 2.10(ii) is

$$\frac{\tau}{\kappa} = \left(\frac{\kappa'}{\tau\kappa^2} \right)'.$$

Set $u = \kappa^{-2}$. Since $\tau = A\kappa$,

$$\frac{\kappa'}{\tau\kappa^2} = \frac{\kappa'}{A\kappa^3} = -\frac{1}{2A}u'.$$

The spherical-curve criterion becomes

$$A = -\frac{1}{2A}u'',$$

or

$$u'' = -2A^2.$$

Therefore

$$u(s) = -A^2s^2 + Bs + C,$$

and hence

$$\boxed{\kappa^2(s) = (-A^2s^2 + Bs + C)^{-1}.}$$

Each step is reversible, proving the converse as well.

Problem 16, continued. *Prove that a spherical slope line through a point on the equator can never reach the north pole. Show that it ends at a point where it meets a small circle about the north pole orthogonally.*

After translating and rescaling, take the unit sphere and let the fixed slope direction v be its polar axis. Put

$$\alpha = \langle T, v \rangle.$$

This is constant. If the curve starts on the equator at $s = 0$, its height is

$$z(s) = \langle c(s), v \rangle = \alpha s.$$

If the curve reached the north pole, its tangent there would be orthogonal to v , forcing $\alpha = 0$. But then $z \equiv 0$, so the curve would remain on the equator. Thus it cannot reach the pole.

For $\alpha > 0$, write

$$c(s) = \left(\sqrt{1 - \alpha^2 s^2} \cos \theta(s), \sqrt{1 - \alpha^2 s^2} \sin \theta(s), \alpha s \right).$$

The unit-speed condition gives

$$\theta'(s)^2 = \frac{1 - \alpha^2 - \alpha^2 s^2}{(1 - \alpha^2 s^2)^2}.$$

Hence

$$\alpha^2 s^2 \leq 1 - \alpha^2.$$

At the endpoint equality holds and $\theta' = 0$. The tangent then has no component along the corresponding small circle, so the curve meets that circle orthogonally.

Problem 18. Show that the Frenet equations for a space curve are equivalent to the Darboux equations $e'_i = D \times e_i$ for $i = 1, 2, 3$, where $D = \tau e_1 + \kappa e_3$ is the Darboux rotation vector.

Solution. Let

$$D = \tau e_1 + \kappa e_3.$$

Using the orientation relations

$$e_1 \times e_2 = e_3, \quad e_2 \times e_3 = e_1, \quad e_3 \times e_1 = e_2,$$

we obtain

$$D \times e_1 = \kappa e_2,$$

$$D \times e_2 = -\kappa e_1 + \tau e_3,$$

$$D \times e_3 = -\tau e_2.$$

These are exactly the Frenet equations

$$e'_1 = \kappa e_2, \quad e'_2 = -\kappa e_1 + \tau e_3, \quad e'_3 = -\tau e_2.$$

Thus the Frenet equations are equivalent to

$e'_i = D \times e_i, \quad i = 1, 2, 3.$

Problem 20. *Prove that c is a helix if and only if its Darboux vector D is constant. Prove also that c is a slope line if and only if $D/\|D\|$ is constant.*

Solution. Differentiating $D = \tau e_1 + \kappa e_3$ gives

$$\begin{aligned} D' &= \tau' e_1 + \tau \kappa e_2 + \kappa' e_3 - \kappa \tau e_2 \\ &= \tau' e_1 + \kappa' e_3. \end{aligned}$$

Therefore

$$D' = 0 \quad \Longleftrightarrow \quad \kappa' = \tau' = 0.$$

A Frenet curve has constant curvature and torsion precisely when it is a circular helix, including the planar circle as the case $\tau = 0$. Hence

$$\boxed{c \text{ is a helix} \quad \Longleftrightarrow \quad D \text{ is constant.}}$$

Since

$$\frac{D}{\|D\|} = \frac{\tau e_1 + \kappa e_3}{\sqrt{\kappa^2 + \tau^2}},$$

a constant ratio τ/κ makes this a fixed vector. Conversely, if $D/\|D\|$ is fixed, it is everywhere orthogonal to e_2 ; Theorem 2.11 then implies that c is a slope line. Thus

$$\boxed{c \text{ is a slope line} \quad \Longleftrightarrow \quad \frac{D}{\|D\|} \text{ is constant.}}$$

Problem 23. Let c be a Frenet curve in $\mathbb{R}^n$. Show that

$$\det(c', c'', \dots, c^{(n)}) = \prod_{i=1}^{n-1} \kappa_i^{n-i}.$$

Solution. With respect to the positively oriented Frenet frame,

$$c' = e_1.$$

Repeated differentiation of the Frenet equations shows inductively that

$$c^{(j)} = (\kappa_1 \kappa_2 \cdots \kappa_{j-1}) e_j + \sum_{i < j} a_{ji} e_i.$$

Thus the coordinate matrix of

$$c', c'', \dots, c^{(n)}$$

in the Frenet frame is triangular. Its diagonal entries are

$$1, \quad \kappa_1, \quad \kappa_1 \kappa_2, \quad \dots, \quad \kappa_1 \kappa_2 \cdots \kappa_{n-1}.$$

Since the Frenet frame has determinant 1,

$$\begin{aligned} \det(c', c'', \dots, c^{(n)}) &= \prod_{j=2}^n (\kappa_1 \cdots \kappa_{j-1}) \\ &= \prod_{i=1}^{n-1} \kappa_i^{n-i}. \end{aligned}$$